# Dimitrios Christos Kyriakou

**Electrical & Computer Engineer · Biomedical Signal Processing & ML**

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## PROFILE

Electrical & Computer Engineer (AUTh, 2026) shaped by research abroad: Texas A&M University (USA, 2024) and Fudan University (China, GRIPS 2025). Award-winning finalist at Texas A&M's Invent for the Planet (2022). Diploma thesis graded 10/10: wrist pulse-rate variability adds about 0.30 ROC-AUC to Parkinson's detection beyond age and sex. European bronze medallist and 5-time Greek champion in Taekwon-Do; Greek offshore sailing champion. Outside the lab, builds 3D-tour and web products for real businesses.

## EDUCATION

**Integrated Master of Engineering (MEng), Electrical & Computer Engineering** — Aristotle University of Thessaloniki 2021 – 2026

- GPA 7.42 / 10 (weighted) · 302 ECTS · EQF Level 7 · focus on signals, data science and machine learning
- Top grades: Data Analysis 10, Biomedical Technology 10, Measurement Systems 10, Electronics II 10, Neural Networks & Deep Learning 9.6, Computational Intelligence 9.5, Computer Networks I 9, Estimation & Detection Theory 8.5

**Diploma thesis** — Digital Biomarkers from Wrist Photoplethysmography for Parkinson's Disease Discrimination Using Machine Learning Grade 10 / 10

Supervisor: Prof. Leontios Hadjileontiadis · 30 ECTS

- End-to-end Python pipeline on free-living PPG + accelerometer data from the PPMI cohort (344 participants: PD, prodromal, healthy controls).
- 4-stage signal-quality gate (motion, waveform shape, template correlation ≥ 0.70, post-motion buffer): 249.3 M candidate 30 s windows → 101.3 M clean windows.
- 27 pulse-rate-variability features (time, frequency, nonlinear) at 30 s and 5 min scales, day and night → 594 features per person.
- Logistic Regression, Random Forest, XGBoost and SVM with participant-level nested cross-validation and Nadeau–Bengio corrected tests.
- **Result:** PRV features improved PD-vs-control ROC-AUC by 0.298–0.323 over age + sex in all four models ($p < 0.05$), robust across sensitivity analyses.

## INTERNATIONAL RESEARCH & PROGRAMS

**GRIPS 2025 Research Intern** — Fudan University, Shanghai, China Summer 2025

- Selected for the Global Research Immersion Program for Young Scientists, with a recommendation from Prof. Gerard L. Coté (Texas A&M).
- Neural and Intelligent Engineering Centre, ISTBI, under Prof. Shouyan Wang, Vice Dean of the institute.
- In a 5-person international team, analysed clinical local field potential (LFP) recordings from a Parkinson's patient, with deep brain stimulation off and on.
- Built the Python pipeline: beta oscillations (13–30 Hz) via Welch PSD, spectrograms and FOOOF; 130 Hz DBS artifacts removed with a predict–blank–interpolate method.
- **Gained:** neural signal processing (time-frequency, connectivity, entropy), closed-loop neuromodulation system design, research in a multinational team.

**Undergraduate Research Intern** — Optical Biosensing Laboratory, Texas A&M University, College Station, USA Summer 2024

- Paired with a PhD student in Prof. Gerard L. Coté's lab (Center for Remote Health Technologies and Systems) on wearable multiwavelength optical sensors.
- Ran an IRB-approved human-subjects study: 2 PPG devices at chest, forehead and finger, 3 contact pressures, validated against a Masimo oximeter.
- Wrote the Python processing and visualisation (perfusion-index heatmaps, 3D comparisons); presented the work as first author of a research poster.
- **Gained:** clinical study protocols and sensor validation, plus a recommendation letter stating I stood out "compared to other students at his educational level".

**Invent for the Planet 2022** — Texas A&M University, USA March – April 2022

- 48-hour global design competition: build a working prototype and pitch it to judges.
- Team BIO-thess (AUTh) was chosen as a top finalist and flown to College Station, Texas, for the live final, with travel funded by Texas A&M.
- Won the **People's Choice award among 25 teams** ($1,000 team prize) for "Energy from Home to Home", an anaerobic digester turning household organic waste into gas for building heating.
- **Gained:** rapid engineering design, prototyping and pitching to an international jury.

**Mentor, JA Europe Foodathon 2023** — Junior Achievement Europe Oct – Nov 2023

- Volunteered as a mentor for Foodathon 2023 (16 Oct – 10 Nov 2023).

## ATHLETICS & DISTINCTIONS

- **Taekwon-Do (ITF):** bronze medal, European Championship 2018 · 5-time Greek National Champion
- **Offshore sailing:** 1st place, Panhellenic Offshore Sailing Championship (ORC B class), 2023
- **Academic:** diploma thesis graded 10/10 · People's Choice Winner, Invent for the Planet 2022

Years of competitive sport alongside engineering: daily training, performing under pressure, and crew work offshore.

## BEYOND THE LAB: VENTURES & SIDE PROJECTS

- **Founder, ART.IO — 3D virtual tours** (artio3d.github.io) — side venture; built the product and its website: interactive 3D walkthroughs with a dollhouse view for real estate, hospitality, event venues, architecture and yachting.
- **Elysium Retreat** (elysiumretreat.gr) — own boutique short-term rental in central Thessaloniki, run since 2022. Designed and built the website (photo tour, amenities, parking guide, booking link); handle guest communication and operations.

## SKILLS

- **Signal processing:** biomedical signals (PPG, accelerometry, LFP), filtering, signal-quality indices, artifact removal, Welch PSD, spectrograms, FOOOF, HRV/PRV analysis
- **Machine learning & statistics:** nested cross-validation, feature selection, Logistic Regression, Random Forest, XGBoost, SVM, neural networks, ANCOVA, ICC, corrected resampled t-tests
- **Programming:** Python (NumPy, Pandas, SciPy, scikit-learn, statsmodels, Matplotlib), Jupyter, SQL / PostgreSQL, data structures & algorithms
- **Web & tools:** HTML/CSS/JS, Decap CMS, Netlify, SEO and web performance, AI assistants (Claude) for business workflows

## CERTIFICATIONS & LANGUAGES

- Data Analysis with Python — IBM / Coursera, 2024
- Reference letter from Prof. Gerard L. Coté (Texas A&M) available on request

- **Greek:** native
- **English:** C2 — Michigan State University Certificate of English Language Proficiency (MSU-CELP)
- **Russian:** A1 — Saint Petersburg State University (TEL/A1), 2023